Who Have Become the Next Generation Pioneers?
－Co-Author Network Analyses in Japanese Academic Societies－

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Abstract

This paper presents co-author network analyses of academic papers in computer science stored in the JST (Japanese Science and Technology agency) database, which contains the titles, unique author IDs with on 286,126 papers and 475,258 authors during 2002 and 2012. We focus on the growth of younger researchers. For the purpose, we set three year moving windows in the database. Next, we construct the co-author network in every three year, measure the betweenness centrality of the network in every three years, and identify the top 0.5% authorities in the years. Based on the information, we search for co-authors of the former authorities, who have become the 0.5% authorities during 2009 and 2012, but have not been contained within the 25% authorities during 2002 and 2005. From the analyses, we have found the 127 pioneering researchers, who have become the top 0.5% authorities.

Keyword: JST Academic Paper database, Co-author network, Network Analysis, Betweenness Centrality Measure

Introduction

Network analysis of academic papers have been carried out for many years [1], [2]. In this paper, we present co-author network analyses of academic papers in computer science field stored in the JST (Japanese Science and Technology agency) database, which contains the titles, unique author IDs with on 286,126 papers and 475,258 authors during 2002 and 2012. We focus on the growth of younger researchers in the sense of the number of published papers. By the growth of researchers, we mean the betweennes centralities of the researcher becomes larger. In this paper, we focus on the extraction of the patterns of the growth.

Analysis Method

In the study, we follow the procedure:

1) We divide the original database during 2002 and 2012 into every three years, then get the eight dataset;
2) We construct the co-author network by giving an undirected arc between every co-author in each paper.
3) We measure the betweenness centrality of every co-author, then identify the top 0.5% authorities in each dataset.
4) Based on the information, we search for co-authors of the former authorities, who have become the 0.5% authorities during 2009 and 2012, but have not been contained within 25% authorities during 2002 and 2005.

Results and Discussion

The summary of the eight data set is shown in Table 1.

Table 1. The summary of the eight data set

	2002-2005	2003-2006	2004-2007	2005-2008	2006-2009	2007-2010	2008-2011	2009-2012
Number of Papers	59201	83255	85923	91153	95479	98113	101646	98976
Number of Authors	108299	149034	157851	169430	178679	188035	199711	202226
Network density	4.31E-5	3.26E-5	3.23E-5	3.13E-5	3.04E-5	2.91E-5	2.73E-5	2.67E-5

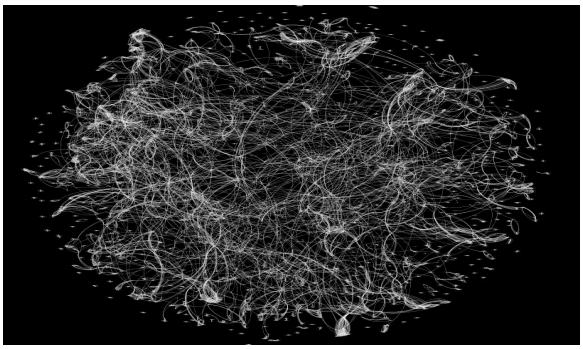


Figure 1. Network of researchers in computer science field

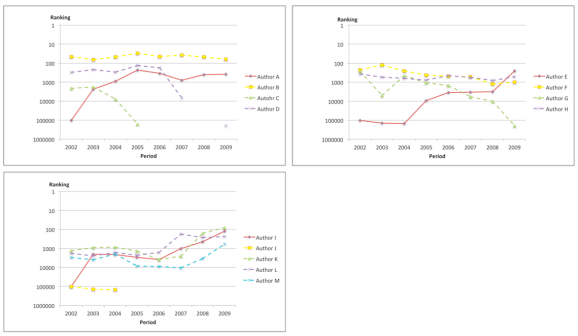


Figure 2. Growth pattern of the three types

And, The co-author network of computer science field is shown in Figure 1.
From the analyses, we have found the 127 pioneering researchers, who have become the top 0.5% authorities. The growth patterns of them are classified into the three types shown in Figure 2.

The three patterns are:

1) Independent researchers without co-author relationship
2) Independent researchers with co-author relationship
3) Group of continuous co-author relationships

The difference comes from the characteristics of the academic subfield. Further detailed analysis will be carried out and presented elsewhere.

References

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